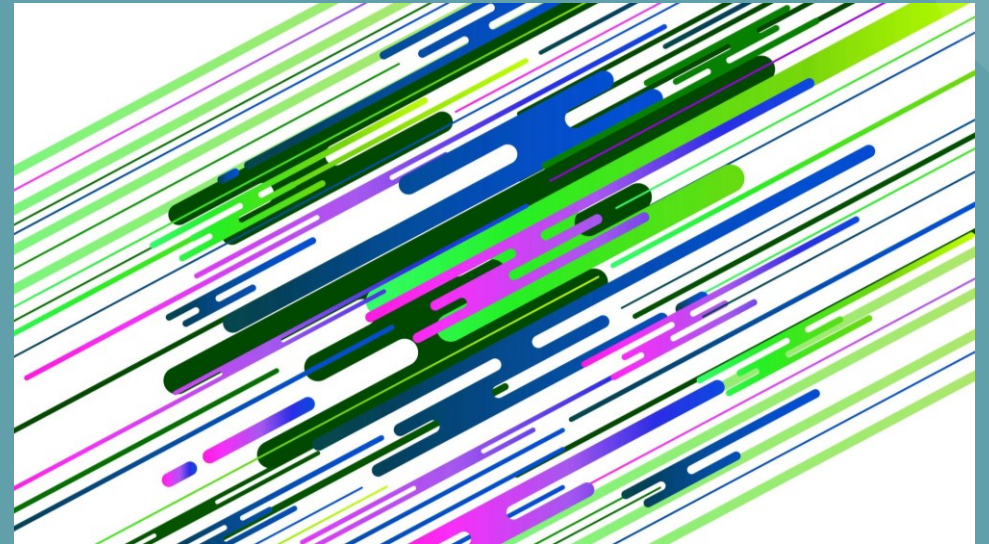


UNIT – FIVE INVENTORY MANAGEMENT



Inventory Classification

Inventory classification is the process of categorizing inventory items based on specific criteria to facilitate efficient inventory management and decision-making.

It involves organizing inventory items into distinct groups or categories to better understand their characteristics, behaviors, and value within the overall inventory.

The purpose of inventory classification is to optimize inventory control by identifying high-value items, managing stock levels effectively, mitigating risks associated with slow-moving or obsolete items, and allocating resources efficiently.

Classification by Inventory Type

Raw Materials: Basic materials used in production.

Work in Progress (WIP): Items in the production process but not yet finished.

Finished Goods: Completed items ready for sale.

MRO (Maintenance, Repair, and Operations): Items used to support operations but not directly incorporated into products.

Obsolete or Discontinued: Items that are no longer used or sellable.

Classification by Contribution to Sales

Classifying inventory based on the importance of items in generating sales.

Types:

- A-Class: High-value items with a significant contribution to sales and profits.
- B-Class: Moderate-value items with a moderate contribution to sales and profits.
- C-Class: Low-value items with a relatively small contribution to sales and profits.

Classification by Frequency of Sales

Classifying inventory based on the frequency of sales for each item.

Types:

- Fast-Moving: Items with high sales frequency, often associated with high customer demand.
- Slow-Moving: Items with low sales frequency, indicating lower customer demand or longer shelf life.
- Non-Moving: Items with no sales over a specified period, typically indicating obsolete or discontinued items.

Classification by Value of Inventory

Classifying inventory based on the monetary value of each item.

Types:

- High-Value: Items with a high monetary value, usually representing a significant portion of the total inventory value.
- Medium-Value: Items with a moderate monetary value, contributing a moderate share to the total inventory value.
- Low-Value: Items with a low monetary value, individually representing a small portion of the total inventory value.

Benefits of Inventory Classification

- Improved inventory control and organization.
- Optimal allocation of resources and capital.
- Effective risk management and reduction of obsolete inventory.
- Streamlined decision-making and strategic planning.
- Enhanced customer satisfaction and service levels.

Costs Associated with Inventory

Inventory costs refer to the various expenses and expenditures incurred in relation to holding and managing inventory.

Purchase Costs

Purchase costs are the expenses associated with acquiring inventory from suppliers or manufacturers.

- Components of Purchase Costs:
 - Unit cost: The price paid per unit of inventory.
 - Freight and transportation costs: Expenses incurred to transport inventory to the warehouse or point of sale.
 - Import duties or customs charges: Taxes or fees imposed on imported goods.
 - Discounts or rebates: Reductions in the purchase price based on negotiated agreements or bulk purchases.

Carrying Costs or Holding Costs

Carrying costs, also known as holding costs, refer to the expenses incurred to hold and maintain inventory.

Components of Carrying Costs:

- Storage costs: Expenses related to renting or owning warehouse space, including rent, utilities, insurance, and security.
- Inventory holding costs: Expenses for handling, inspecting, and storing inventory, such as labor, equipment, and maintenance costs.
- Opportunity cost of capital: The cost of tying up capital in inventory instead of investing it elsewhere.
- Depreciation and obsolescence: Losses due to the deterioration, spoilage, or obsolescence of inventory.
- Insurance and taxes: Premiums paid for insuring inventory against damage or loss, as well as property taxes on stored inventory.

Ordering Costs

Ordering costs are the expenses incurred in the process of replenishing inventory.

Components of Ordering Costs:

Purchase order processing: Administrative costs associated with creating and processing purchase orders.

Supplier communication and coordination costs: Expenses related to communicating with suppliers, negotiating terms, and managing supplier relationships.

Quality control and inspection costs: Expenses incurred to ensure the received inventory meets quality standards.

Receiving and inspection costs: Labor and equipment costs involved in receiving, inspecting, and verifying the accuracy of delivered inventory.

Shortage Costs

Shortage costs are the expenses incurred due to insufficient inventory levels to meet customer demand.

Components of Shortage Costs:

Lost sales and revenue: Potential revenue not realized due to stockouts and unfulfilled customer orders.

Backordering costs: Expenses associated with processing and managing backorders, including additional administrative and customer service efforts.

Expedited shipping or production costs: Additional charges incurred for rush shipping or expedited production to fulfill urgent orders.

Damaged customer relationships: Costs associated with dissatisfied customers, including potential loss of future business and negative word-of-mouth.

Economic Order Quantity (EOQ)

Economic Order Quantity (EOQ) is a calculation method used to find the order quantity that minimizes the total cost of inventory management.

It considers factors such as annual demand, ordering costs, and holding costs per unit to determine the optimal order quantity.

The goal of EOQ is to find the balance between holding excess inventory (incurring carrying costs) and frequent ordering (incurring ordering costs) to achieve cost-effective inventory control.

EOQ Equation

$$EOQ = \sqrt{\frac{2AO}{C}}$$

Where,

A = Annual requirement

O = Ordering costs per order

C = Carrying costs per unit

The background of the slide is a dark, textured surface with a marbled or topographical pattern. It features intricate, swirling lines in shades of dark blue, grey, and black, creating a complex, organic texture.

Inventory Models with Deterministic Demand

Deterministic Single Item Inventory Models

Economic order quantity

Economic production model

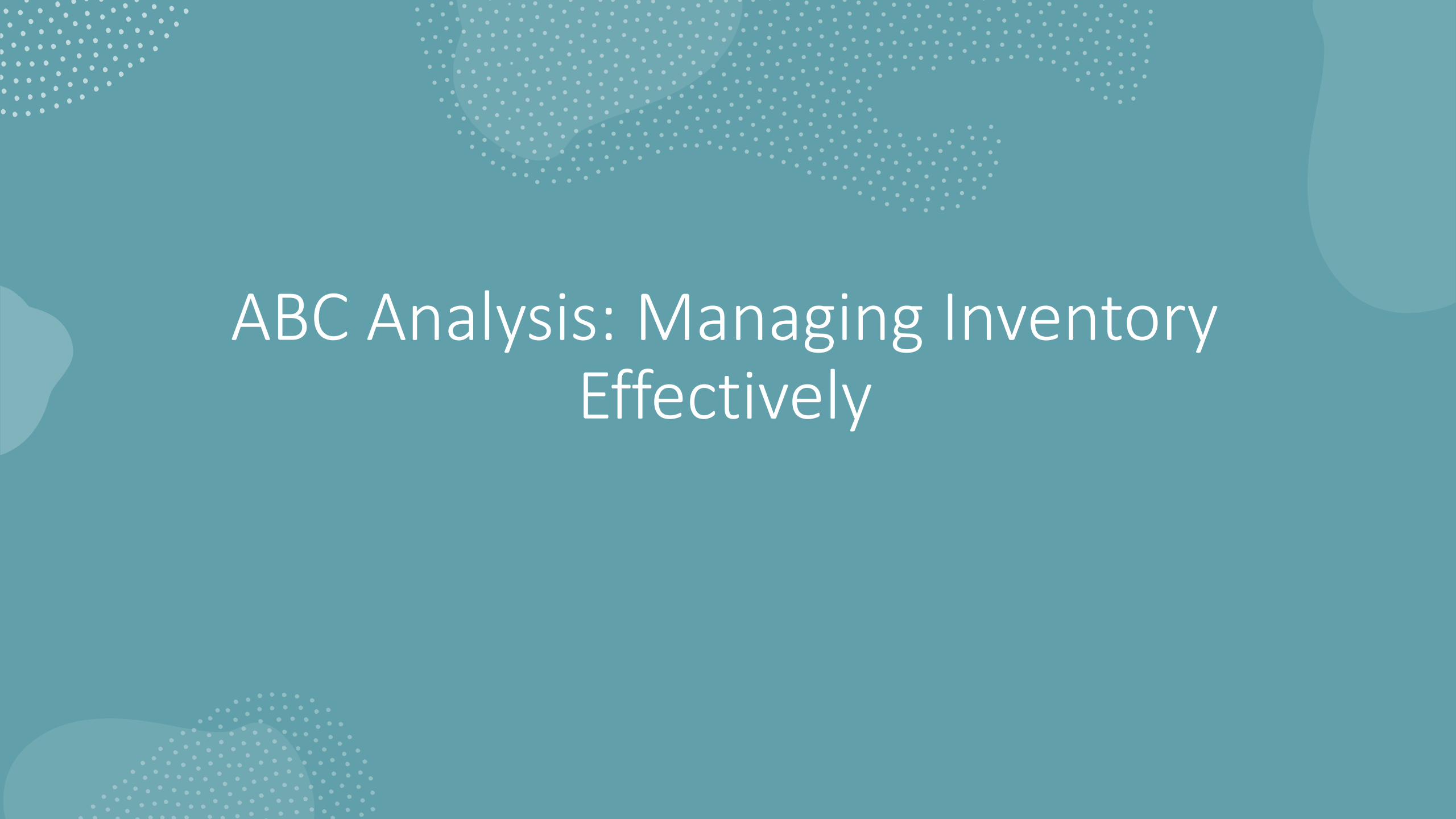
Price discount model

Dynamic demand model

Deterministic Multi- item Inventory Models

Unknown cost
structure model

Known cost
structure model



ABC Analysis: Managing Inventory Effectively

ABC Analysis



A technique for categorizing items based on their importance in inventory management.



Enables effective allocation of resources and prioritization of efforts.



Classifies items into three categories: A, B, and C.



Categories in ABC Analysis – (I)

Category A (High-value items)

- Items with the highest value or impact on the business.
 - Typically, constitutes a small percentage of total items but contributes to a significant portion of the value.
 - Requires close monitoring and frequent review.
-

Categories in ABC Analysis - (II)

A thick yellow horizontal bar spans the width of the slide, with a vertical yellow bar of the same thickness extending downwards from its right end.

Category B (Moderate-value items)

- Items with moderate value or impact.
- Represents a moderate percentage of total items and contributes to a moderate portion of the value.
- Requires regular monitoring and periodic review.

Categories in ABC Analysis – (III)

A thick yellow horizontal bar spans the width of the slide, with a vertical yellow bar of the same thickness extending downwards from its right end.

Category C (Low-value items)

- Items with low value or impact.
- Makes up most total items but contributes to a small portion of the value.
- Requires minimal monitoring and infrequent review.

Example: ABC Analysis in Retail Store

Item	Annual Revenue (Rs,)	Annual Sales Quantity
Item A (Category A)	Rs. 50,000	100 units
Item B (Category B)	Rs. 20,000	200 units
Item C (Category B)	Rs. 15,000	300 units
Item D (Category C)	Rs. 5,000	500 units
Item E (Category C)	Rs. 2,000	800 units



Example: ABC Analysis in Retail Store (contd.)

Category A (High-value items):

- Item A: Annual Revenue - Rs.50,000 (highest in the store)
- 100 units sold annually
- Requires close monitoring and frequent review due to its significant value.

Example:
ABC Analysis
in Retail
Store
(contd.)

Category B (Moderate-value items):

- Item B: Annual Revenue - Rs.20,000
- 200 units sold annually
- Item C: Annual Revenue - Rs.15,000
- 300 units sold annually
- Both items require regular monitoring and periodic review.



Example: ABC Analysis in Retail Store (contd.)

Category C (Low-value items):

- Item D: Annual Revenue - Rs.5,000
- 500 units sold annually
- Item E: Annual Revenue - Rs.2,000 (lowest in the store)
- 800 units sold annually
- Minimal monitoring and infrequent review required.